



```
In [1]: import numpy as np
```

#Seasons

```
Seasons = ["2010", "2011", "2012", "2013", "2014", "2015", "2016", "2017", "2018", "2019"]  
Sdict = {"2010":0, "2011":1, "2012":2, "2013":3, "2014":4, "2015":5, "2016":6, "2017":7, "2018":8, "2019":9}
```

#Players

```
Players = ["Sachin", "Rahul", "Smith", "Sami", "Pollard", "Morris", "Samson", "Dhoni", "Kohli", "Sky"]  
Pdict = {"Sachin":0, "Rahul":1, "Smith":2, "Sami":3, "Pollard":4, "Morris":5, "Samson":6, "Dhoni":7, "Kohli":8, "Sky":9}
```

#Salaries

```
Sachin_Salary = [15946875, 17718750, 19490625, 21262500, 23034375, 24806250, 25244490, 26687500, 28125000, 29562500]  
Rahul_Salary = [12000000, 12744189, 13488377, 14232567, 14976754, 16324500, 18038573, 19752646, 21466719, 23180792]  
Smith_Salary = [4621800, 5828090, 13041250, 14410581, 15779912, 14500000, 16022500, 17545000, 19067500, 20590000]  
Sami_Salary = [3713640, 4694041, 13041250, 14410581, 15779912, 17149243, 18518574, 19887905, 21257236, 22626567]  
Pollard_Salary = [4493160, 4806720, 6061274, 13758000, 15202590, 16647180, 18091770, 19536360, 20980950, 22425540]  
Morris_Salary = [3348000, 4235220, 12455000, 14410581, 15779912, 14500000, 16022500, 17545000, 19067500, 20590000]  
Samson_Salary = [3144240, 3380160, 3615960, 4574189, 13520500, 14940153, 16359805, 17779458, 19199111, 20618764]  
Dhoni_Salary = [0, 0, 4171200, 4484040, 4796880, 6053663, 15506632, 16669630, 17832627, 18995625]  
Kohli_Salary = [0, 0, 0, 4822800, 5184480, 5546160, 6993708, 16402500, 17632688, 18862876]  
Sky_Salary = [3031920, 3841443, 13041250, 14410581, 15779912, 14200000, 15691000, 17182000, 18673000, 20164000]
```

#Matrix

```
Salary = np.array([Sachin_Salary, Rahul_Salary, Smith_Salary, Sami_Salary, Pollard_Salary, Morris_Salary, Samson_Salary, Dhoni_Salary, Kohli_Salary, Sky_Salary])
```

#Games

```
Sachin_G = [80, 77, 82, 82, 73, 82, 58, 78, 6, 35]  
Rahul_G = [82, 57, 82, 79, 76, 72, 60, 72, 79, 80]  
Smith_G = [79, 78, 75, 81, 76, 79, 62, 76, 77, 69]  
Sami_G = [80, 65, 77, 66, 69, 77, 55, 67, 77, 40]  
Pollard_G = [82, 82, 82, 79, 82, 78, 54, 76, 71, 41]  
Morris_G = [70, 69, 67, 77, 70, 77, 57, 74, 79, 44]  
Samson_G = [78, 64, 80, 78, 45, 80, 60, 70, 62, 82]  
Dhoni_G = [35, 35, 80, 74, 82, 78, 66, 81, 81, 27]  
Kohli_G = [40, 40, 40, 81, 78, 81, 39, 0, 10, 51]  
Sky_G = [75, 51, 51, 79, 77, 76, 49, 69, 54, 62]
```

#Matrix

```
Games = np.array([Sachin_G, Rahul_G, Smith_G, Sami_G, Pollard_G, Morris_G, Samson_G, Dhoni_G, Kohli_G, Sky_G])
```

#Points

```
Sachin_PTS = [2832, 2430, 2323, 2201, 1970, 2078, 1616, 2133, 83, 782]  
Rahul_PTS = [1653, 1426, 1779, 1688, 1619, 1312, 1129, 1170, 1245, 1154]  
Smith_PTS = [2478, 2132, 2250, 2304, 2258, 2111, 1683, 2036, 2089, 1743]  
Sami_PTS = [2122, 1881, 1978, 1504, 1943, 1970, 1245, 1920, 2112, 966]  
Pollard_PTS = [1292, 1443, 1695, 1624, 1503, 1784, 1113, 1296, 1297, 646]  
Morris_PTS = [1572, 1561, 1496, 1746, 1678, 1438, 1025, 1232, 1281, 928]  
Samson_PTS = [1258, 1104, 1684, 1781, 841, 1268, 1189, 1186, 1185, 1564]  
Dhoni_PTS = [903, 903, 1624, 1871, 2472, 2161, 1850, 2280, 2593, 686]  
Kohli_PTS = [597, 597, 597, 1361, 1619, 2026, 852, 0, 159, 904]  
Sky_PTS = [2040, 1397, 1254, 2386, 2045, 1941, 1082, 1463, 1028, 1331]
```

#Matrix

```
Points = np.array([Sachin_PTS, Rahul_PTS, Smith_PTS, Sami_PTS, Pollard_PTS, Morris_PTS, Samson_PTS, Dhoni_PTS, Kohli_PTS, Sky_PTS])
```

```
In [2]: Salary
```

```
Out[2]: array([[15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
                25244493, 27849149, 30453805, 23500000],
               [12000000, 12744189, 13488377, 14232567, 14976754, 16324500,
                18038573, 19752645, 21466718, 23180790],
               [ 4621800,  5828090, 13041250, 14410581, 15779912, 14500000,
                16022500, 17545000, 19067500, 20644400],
               [ 3713640,  4694041, 13041250, 14410581, 15779912, 17149243,
                18518574, 19450000, 22407474, 22458000],
               [ 4493160,  4806720,  6061274, 13758000, 15202590, 16647180,
                18091770, 19536360, 20513178, 21436271],
               [ 3348000,  4235220, 12455000, 14410581, 15779912, 14500000,
                16022500, 17545000, 19067500, 20644400],
               [ 3144240,  3380160,  3615960,  4574189, 13520500, 14940153,
                16359805, 17779458, 18668431, 20068563],
               [      0,      0,  4171200,  4484040,  4796880,  6053663,
                15506632, 16669630, 17832627, 18995624],
               [      0,      0,      0,  4822800,  5184480,  5546160,
                6993708, 16402500, 17632688, 18862875],
               [ 3031920,  3841443, 13041250, 14410581, 15779912, 14200000,
                15691000, 17182000, 18673000, 15000000]])
```

```
In [5]: Games
```

```
Out[5]: array([[80, 77, 82, 82, 73, 82, 58, 78,  6, 35],
               [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
               [79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
               [80, 65, 77, 66, 69, 77, 55, 67, 77, 40],
               [82, 82, 82, 79, 82, 78, 54, 76, 71, 41],
               [70, 69, 67, 77, 70, 77, 57, 74, 79, 44],
               [78, 64, 80, 78, 45, 80, 60, 70, 62, 82],
               [35, 35, 80, 74, 82, 78, 66, 81, 81, 27],
               [40, 40, 40, 81, 78, 81, 39,  0, 10, 51],
               [75, 51, 51, 79, 77, 76, 49, 69, 54, 62]])
```

```
In [4]: Players
```

```
Out[4]: ['Sachin',
         'Rahul',
         'Smith',
         'Sami',
         'Pollard',
         'Morris',
         'Samson',
         'Dhoni',
         'Kohli',
         'Sky']
```

```
In [6]: Sdict
```

```
Out[6]: {'2010': 0,
        '2011': 1,
        '2012': 2,
        '2013': 3,
        '2014': 4,
        '2015': 5,
        '2016': 6,
        '2017': 7,
        '2018': 8,
        '2019': 9}
```

```
In [7]: Pdict
```

```
Out[7]: {'Sachin': 0,
        'Rahul': 1,
        'Smith': 2,
        'Sami': 3,
        'Pollard': 4,
        'Morris': 5,
        'Samson': 6,
        'Dhoni': 7,
        'Kohli': 8,
        'Sky': 9}
```

```
In [8]: Salary
```

```
Out[8]: array([[15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
                25244493, 27849149, 30453805, 23500000],
               [12000000, 12744189, 13488377, 14232567, 14976754, 16324500,
                18038573, 19752645, 21466718, 23180790],
               [ 4621800,  5828090, 13041250, 14410581, 15779912, 14500000,
                16022500, 17545000, 19067500, 20644400],
               [ 3713640,  4694041, 13041250, 14410581, 15779912, 17149243,
                18518574, 19450000, 22407474, 22458000],
               [ 4493160,  4806720,  6061274, 13758000, 15202590, 16647180,
                18091770, 19536360, 20513178, 21436271],
               [ 3348000,  4235220, 12455000, 14410581, 15779912, 14500000,
                16022500, 17545000, 19067500, 20644400],
               [ 3144240,  3380160,  3615960,  4574189, 13520500, 14940153,
                16359805, 17779458, 18668431, 20068563],
               [      0,      0,  4171200,  4484040,  4796880,  6053663,
                15506632, 16669630, 17832627, 18995624],
               [      0,      0,      0,  4822800,  5184480,  5546160,
                6993708, 16402500, 17632688, 18862875],
               [ 3031920,  3841443, 13041250, 14410581, 15779912, 14200000,
                15691000, 17182000, 18673000, 15000000]])
```

```
In [9]: Salary//Games
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_6132\1634212085.py:1: RuntimeWarning:
divide by zero encountered in floor_divide
Salary//Games
```

```
Out[9]: array([[ 199335,  230113,  237690,  259298,  315539,  302515,  435249,
                 357040, 5075634,  671428],
               [ 146341,  223582,  164492,  180159,  197062,  226729,  300642,
                 274342,  271730,  289759],
               [  58503,   74719,  173883,  177908,  207630,  183544,  258427,
                 230855,  247629,  299194],
               [  46420,   72216,  169366,  218342,  228694,  222717,  336701,
                 290298,  291006,  561450],
               [  54794,   58618,   73917,  174151,  185397,  213425,  335032,
                 257057,  288918,  522835],
               [  47828,   61380,  185895,  187150,  225427,  188311,  281096,
                 237094,  241360,  469190],
               [  40310,   52815,   45199,   58643,  300455,  186751,  272663,
                 253992,  301103,  244738],
               [      0,      0,   52140,   60595,   58498,   77611,  234948,
                 205797,  220155,  703541],
               [      0,      0,      0,   59540,   66467,   68471,  179325,
                 0, 1763268,  369860],
               [  40425,   75322,  255710,  182412,  204933,  186842,  320224,
                 249014,  345796,  241935]])
```

```
In [10]: Games[2,:]
```

```
Out[10]: array([79, 78, 75, 81, 76, 79, 62, 76, 77, 69])
```

```
In [11]: Pdict
```

```
Out[11]: {'Sachin': 0,
          'Rahul': 1,
          'Smith': 2,
          'Sami': 3,
          'Pollard': 4,
          'Morris': 5,
          'Samson': 6,
          'Dhoni': 7,
          'Kohli': 8,
          'Sky': 9}
```

```
In [12]: Pdict
```

```
Out[12]: {'Sachin': 0,
          'Rahul': 1,
          'Smith': 2,
          'Sami': 3,
          'Pollard': 4,
          'Morris': 5,
          'Samson': 6,
          'Dhoni': 7,
          'Kohli': 8,
          'Sky': 9}
```

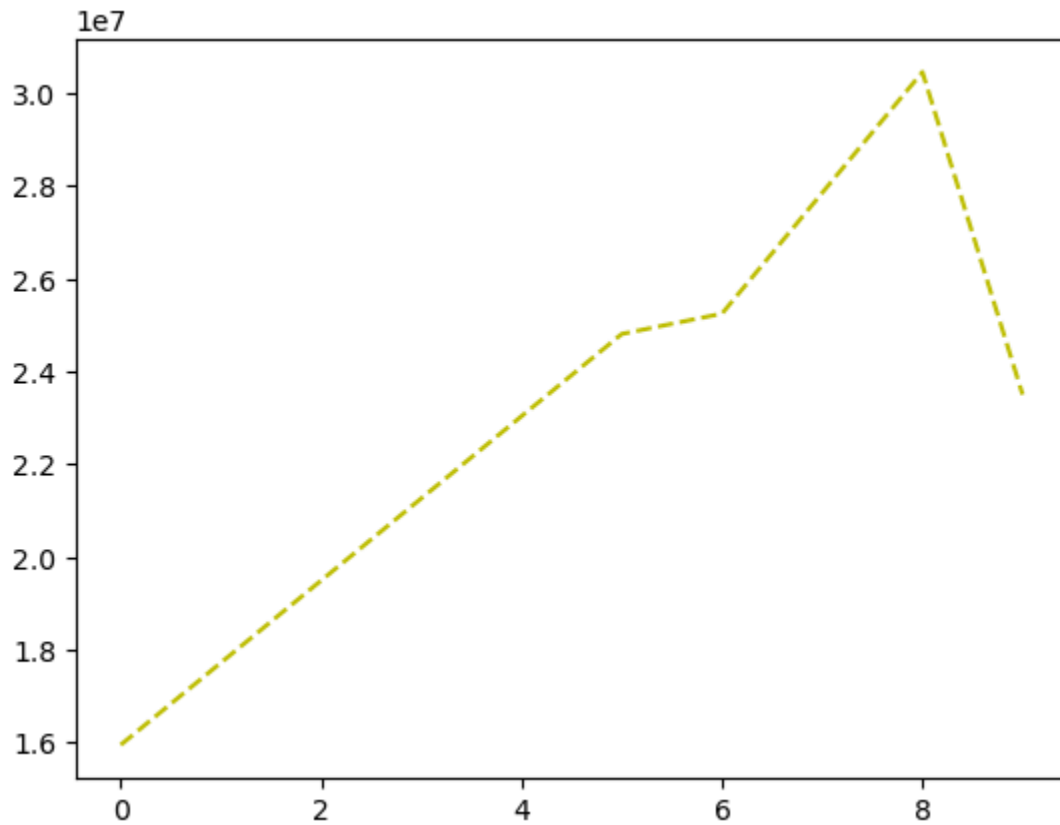
```
In [13]: # to ignore os warning error while visualize the graph
import warnings
```

```
warnings.filterwarnings('ignore')
```

```
In [14]: import matplotlib.pyplot as plt
```

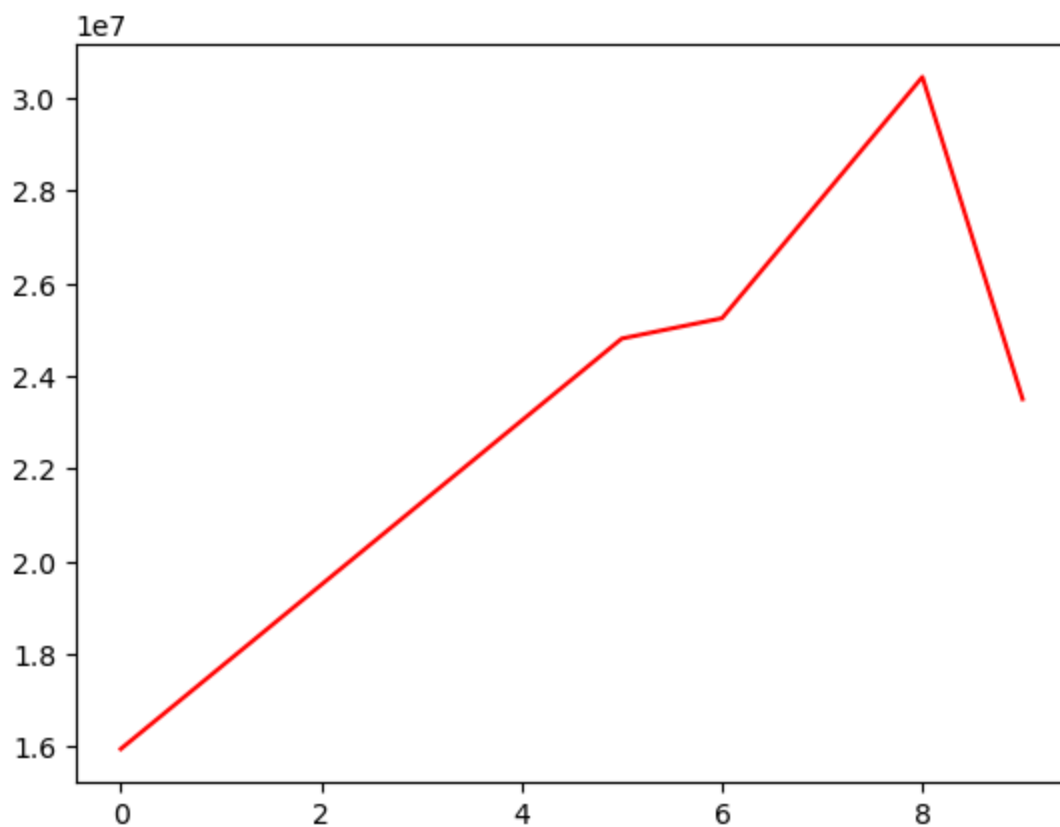
```
In [15]: plt.plot(Salary[0], c='y', ls='--')  
plt.show
```

```
Out[15]: <function matplotlib.pyplot.show(close=None, block=None)>
```



```
In [16]: plt.plot(Salary[0], c='r')
```

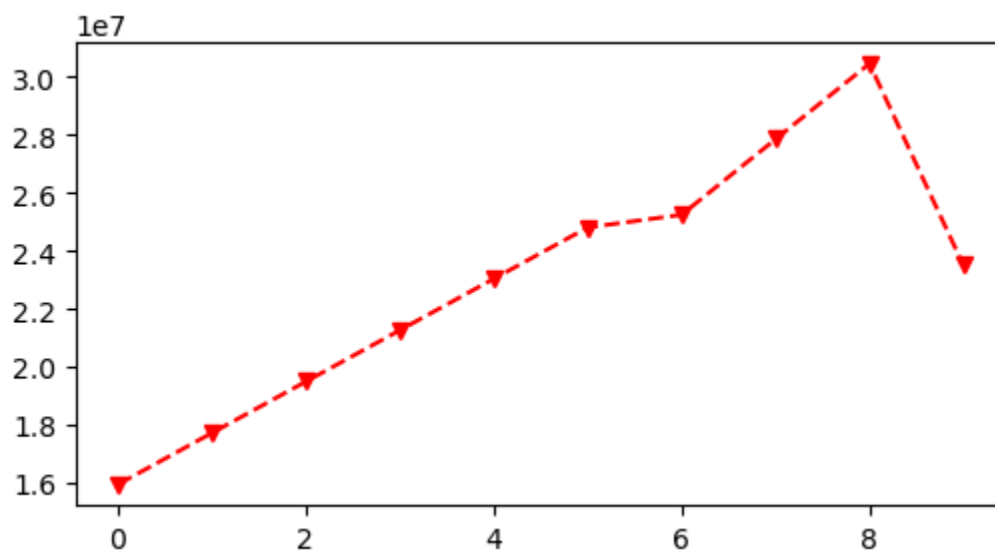
```
Out[16]: [<matplotlib.lines.Line2D at 0x18a3c7879d0>]
```



```
In [17]: %matplotlib inline
plt.rcParams['figure.figsize'] = 6,3
```

```
In [18]: plt.plot(Salary[0], c='r', ls='--', marker='v')
```

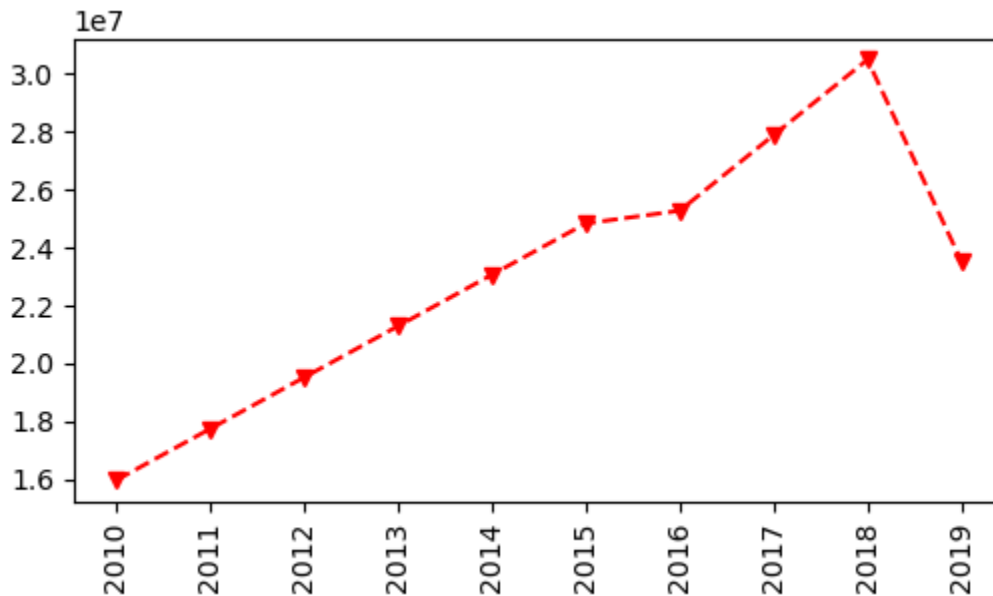
```
Out[18]: [<matplotlib.lines.Line2D at 0x18a3c5dca50>]
```



```
In [19]: plt.plot(Salary[0], c='r', ls='--', marker='v')
plt.xticks(list(range(0,10)),Seasons, rotation='vertical')
```

```
plt.show
```

```
Out[19]: <function matplotlib.pyplot.show(close=None, block=None)>
```

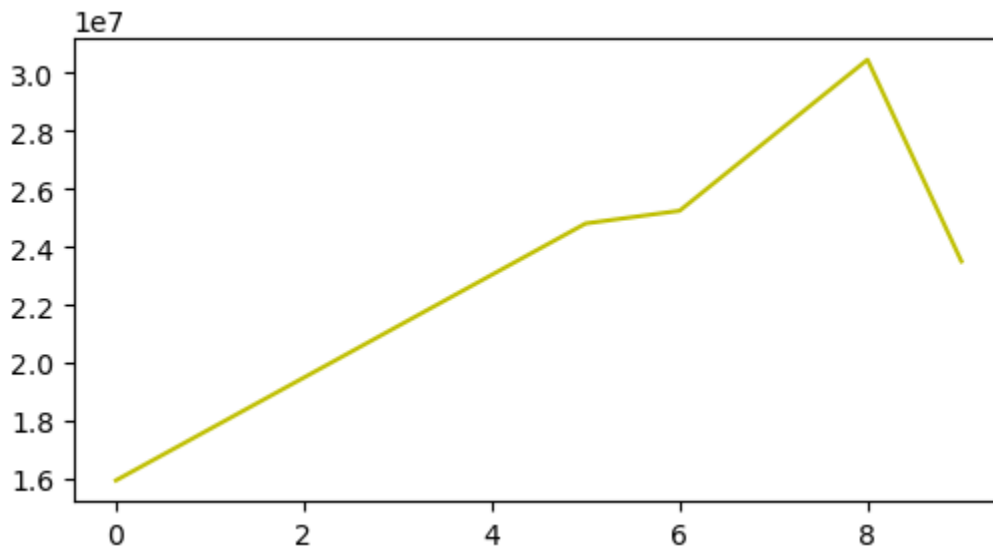


```
In [20]: Games[0]
```

```
Out[20]: array([80, 77, 82, 82, 73, 82, 58, 78, 6, 35])
```

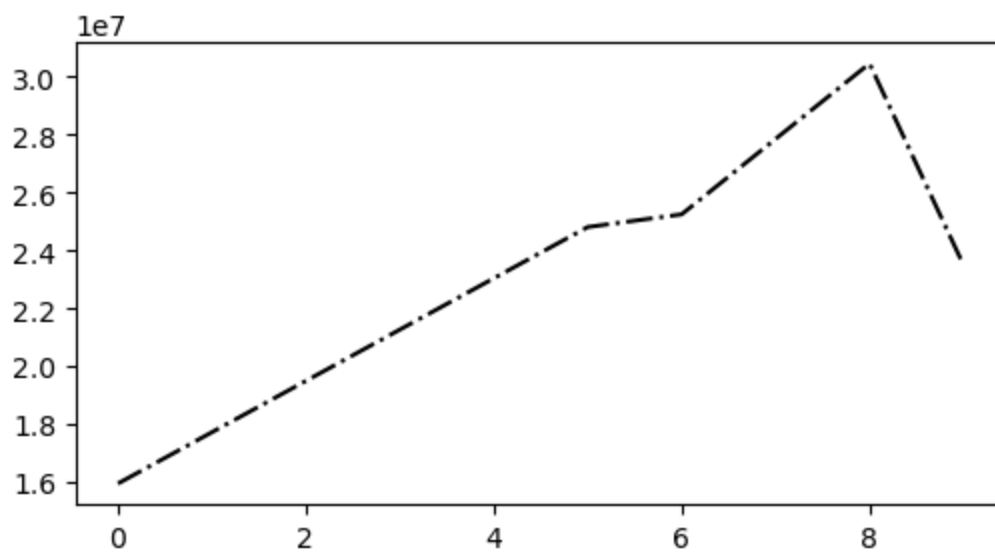
```
In [21]: plt.plot(Salary[0], c='y')
```

```
Out[21]: [<matplotlib.lines.Line2D at 0x18a3c664a50>]
```



```
In [22]: plt.plot(Salary[0], c='k', ls = '-.')
```

```
Out[22]: [<matplotlib.lines.Line2D at 0x18a3c6cafd0>]
```



```
In [23]: plt.plot(Salary[0], c='k', ls = '--', marker = 'o')
```

```
Out[23]: [<matplotlib.lines.Line2D at 0x18a3d86d590>]
```

